

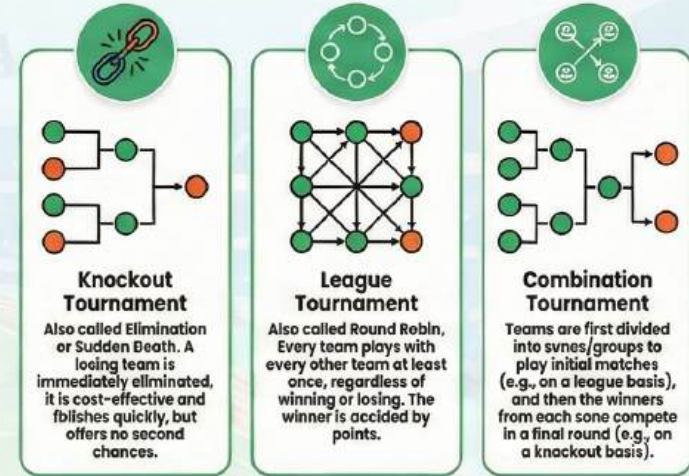
The Ultimate Guide to Managing Sporting Events

The 5 Core Functions of Sports Management



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Tournament Types Explained

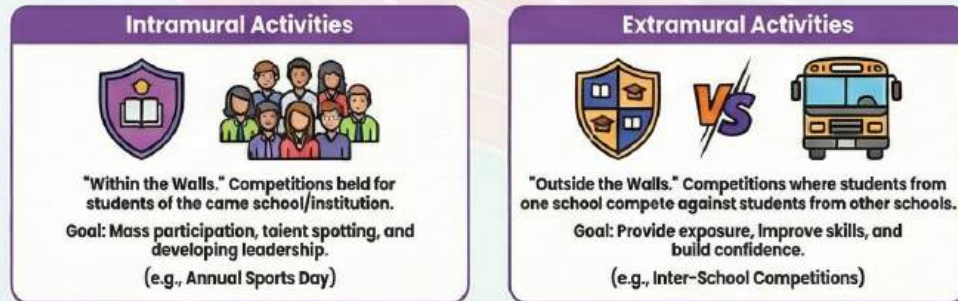


Event Responsibilities: A Three-Phase Approach



Made By Dr Pragya

Intramural vs. Extramural Activities



Community Sports Programs: More Than a Game



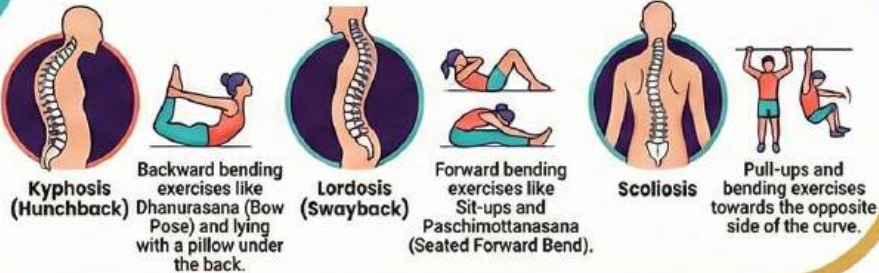
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Children & Women in Sports: A Visual Guide for Class 12 Physical Education

WHO Exercise Guidelines for Lifelong Health



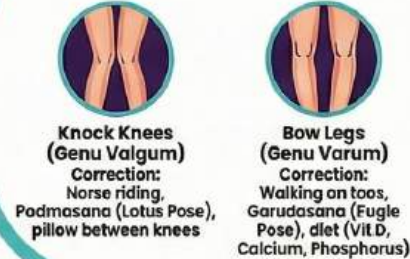
Spinal Deformities



Identifying & Correcting Postural Deformities



Limb Deformities

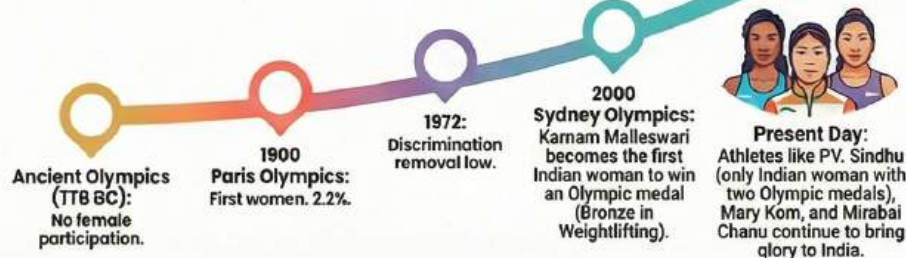


Other Deformities



Women in Sports: Participation & Special Considerations

A Timeline of Participation



Barriers to Participation in India



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1. Osteoporosis
Weak/brittle bones due to low bone density, low estrogen, calcium/Vit D deficiency.

The Female Athlete Triad

A critical syndrome of three interconnected health disorders.

2. Amenorrhea
Absence of menstrual periods for 3+ months; caused by excessive exercise & inadequate calorie intake, hormonal imbalance.

3. Eating Disorders
A mental illness characterized by abnormal eating habits that negatively impact health.

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Yoga as a Preventive Measure: A Visual Guide for Class 12



1. Obesity (मोटापा)

A condition of excessive body fat (BMI 30+), increasing heart disease, high BP, and diabetes risks.



Tadasana
Basic standing pose improving posture, strengthens thighs and ankles, and helps reduce belly fat.



Katichakrasana
Standing twist improving spine flexibility, helps with constipation, and managing obesity.



Pawanmuktasana
Reclining pose aiding digestion, relieves gas and constipation, and helps reduce fat in the abdominal area.



Helasana (Plough Pose)
Fore-body stretch reducing belly fat, improving memory, and managing obesity.

Context Summary

Yoga serves as a powerful holistic practice for maintaining health and preventing diseases. An 'Asana', as defined by sage Patanjali, is a comfortable and steady posture (Sthiram Sukham Asanam). This guide explores asanas as preventive measures for five key lifestyle diseases.

Universal Asana Guidelines



General Benefits:
Improves posture, reduces stress & anxiety, contributes to overall health.



Common Contraindications:
Caution for pregnant women and high BP. Always check specific contraindications.



How to Describe Procedure:
Visualize the posture and describe steps in your own words, avoid rote memorization.

2. Diabetes (मधुमेह)

Chronic disease where the body cannot regulate blood sugar due to insufficient or ineffective insulin.



Bhujangasana (Cobra Pose):
Back-bending pose improving circulation, managing diabetes by stimulating abdominal organs.

Gomubhasana (Cow Face Pose):
Fore-body stretch reducing multiple risks. Leucostol for managing diabetes.



Shalabhasana (Locust Pose):
Lying on stomach, strengthens back, improves stability, and helps remove extra fat from the abdomen and waist.



Kapalbhati (Shali Elling Breath):
Pranayama energizing the body, improving lung capacity, and aiding diabetes management.



3. Asthma (दमा)

Chronic respiratory disease with narrowed, inflamed airways and mucus accumulation, causing breathing difficulties.



Urdhva Hastasana (Upward Salute Pose):
Stretches the body, improves digestion, and enhances lung capacity for better breathing.



Vakrasana (Twisted Pose):
Seated twist improving spinal cord function, controlling diabetes, and beneficial for asthma.



Matsyasana (Fish Pose):
Reclining back-bend stretching neck and shoulders, very helpful for respiratory issues like asthma.



Anulom Vilom (Alternate Nostril Breathing):
Sitting pranayama purifying energy channels, reducing stress, and significantly benefiting respiratory function.



4. Hypertension (उच्च रक्तचाप)

High Blood Pressure condition with consistently high force against artery walls (A normal range is around 120/80 mmHg).



Ardha Halasana (Half Plough Pose):
Lying on back with legs at 90 degrees, improves digestion, tones thighs, and manages hypertension.



Makarasana (Crocodile Pose):
A relaxing pose where one lies on the stomach. It calms the body and making it effective for reducing high blood pressure.



Shavasana
Ultimate relaxation pose. Lying flat on back to calm the mind, improve mental health, and normalize blood circulation, directly combating hypertension.



Sitali Pranayama (Cooling Breath):
A breathing technique where air is inhaled through a U-shaped tongue. It has a cooling, as excellent for lowering blood pressure.

5. Back Pain & Arthritis (पीठ दर्द और गठिया)

Back pain (often muscle strain) and Arthritis (joint inflammation) causing pain, swelling, and stiffness.



Ardha Chakrasana (Half Wheel Pose):
Standing back-bend improving lung capacity, improving spine flexibility, and relieving back pain.



Ustrasana (Camel Pose):
Kneeling back-bend improving spine flexibility, relieving lower back pain, and reducing stomach fat.



Sural Matsyasana (Easy Fish Pose):
Simpler fish pose beneficial for spinal problems, slip disc, and relieving back pain.



Bhadrasana (Gracious Pose):
Seated pose strengthening the spine, improving digestion, and relieving back pain and arthritis by stretching buttocks and ankles.

Unlocking Potential: A Guide to Sports for Children With Special Needs (CWSN)

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FOUNDATIONAL CONCEPTS



CWSN: Children With Special Needs
This term refers to children who are differently-abled or have disabilities, requiring special attention and programs.

A Shift in Language: Around 1980, "differently-abled" was introduced as a respectful alternative to "disabled" or "handicapped" to reduce discrimination.



Disability vs. Disorder
A **disability** is a permanent condition restricting body function (e.g., a lost limb), while a **disorder** is often a temporary, brain-related ailment.

Adaptive Sports: These are sports programs specifically designed for differently-abled individuals, featuring modified rules and equipment to ensure ease of play.



SPECIAL OLYMPICS
"Let the wiv, but if I cannot win, let me be brave in the attempt."

For Intellectual Disabilities:
A global event exclusively for athletes with intellectual disabilities, focusing on participation and effort.

Founded in 1986 by Eunice Kennedy Shriver.



PARALYMPICS
"Spirit in Motion"

The Premier Event for a Range of Disabilities:
The largest multi-sport event for elite athletes with various physical, sensory, and intellectual disabilities.

Inspired by Sir Ludwig Guttmann, officially started in 1980.



DEAFLYMPICS
"International Silent Games."

Elite Competition for Deaf Athletes:
An international multi-sport event for athletes with a hearing loss of at least 66 decibels.

Founded in 1924.

Feature	Special Olympics	Paralympics	Deaflympics
Primary Athletes	Intellectually Disabled	Wide Range of Disabilities	Deaf / Hard of Hearing
Frequency	Every 2 years (Summer/Winter)	Every 4 years (Summer/Winter)	Every 4 years (Summer/Winter)
Governing Body	SGI (Special Olympics Int'l)	IPC (Int'l Paralympic Committee)	ICSD (Int'l Committee of Sports for the Deaf)
Key Focus	Participation, Joy, Courage	Elite Athletic Performance	Elite Athletic Performance
Indian Body	Special Olympics Bharat	Paralympic Committee of India	All India Sports Council of the Deaf

ENSURING FAIR COMPETITION

Classification vs. Divisioning: These are two distinct methods for grouping athletes to ensure fair and equitable competition.



CLASSIFICATION
(in Paralympics)



Primary Basis: Type and impact of the athlete's disability on their sport.
Goal: Group athletes with similar functional limitations.
Process: Involves medical and functional assessments by officials.



DIVISIONING
(in Special Olympics)



Primary Basis: The athlete's skill level, ability, age, and gender.
Goal: Group athletes with similar performance abilities for a fair chance to win.
Process: Based on scores from preliminary events or trials.

INCLUSION AND ACCESSIBILITY IN SPORTS

What is Inclusion?

Creating an environment where people with and without disabilities have equal opportunities to participate in sports together.



Why is Inclusion Important?

It boosts confidence and motor skills for CWSN while teaching empathy, respect, and social skills to all participants.

Strategies to Make Activities Accessible



Medical Check-ups

Understand individual limitations and needs.



Modify Rules

Allow extra attempts, provide longer rest breaks.



Adapt Equipment

Use bright colors for visual impairment or lightweight objects for weaker muscles.



Create a Safe Environment

Ensure proper lighting, avoid loud noises, and have a suitable play area.

THE BENEFITS OF PHYSICAL ACTIVITY FOR CWSN

More Than Just a Game: Sports participation provides powerful holistic benefits for Children With Special Needs.



PHYSICAL BENEFITS

Improves muscle strength, coordination, balance, flexibility, and overall fitness.



SOCIAL BENEFITS

Enhances communication skills, teamwork, and social interaction, leading to new friendships.



PSYCHOLOGICAL & EMOTIONAL BENEFITS

Reduces stress and depression, boosts self-esteem, develops a positive attitude, and builds self-confidence.

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MASTERING SPORTS & NUTRITION: THE ULTIMATE GUIDE FOR STUDENTS

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Comprehensive guide on nutrition, its components, and its crucial role in sports performance for Class 12 Physical Education.

1. The Foundations of Fuel



WHAT IS FOOD?

Any substance consumed for survival. Basic need to stay alive.



WHAT IS NUTRITION?

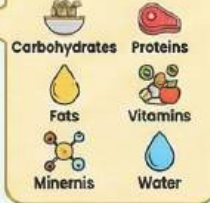
Dynamic process of consuming food to nourish the body, build tissues, provide energy, and stay healthy.



WHAT IS A BALANCED DIET?

Diet containing all essential food groups in correct proportions.

THE 6 PILLARS OF A BALANCED DIET



3. Nutritive vs. Non-Nutritive Components



NUTRITIVE COMPONENTS

Provide energy or calories. Includes carbs, proteins, fats, vitamins, and minerals.



NON-NUTRITIVE COMPONENTS

Do not provide energy, but important functions. Includes fiber, water, color & flavor compounds.



THE POWER OF FIBER (BOUGHES): Crucial for digestion. Lack can lead to constipation.

5. Food Myths Busted



MYTH: Carbohydrates make you fat.



FACT: Carbs are primary energy source. Only excessive consumption leads to fat storage.



MYTH: Eggs raise cholesterol.



FACT: Eggs are high-quality protein, in moderation, they don't significantly raise harmful cholesterol for most.



MYTH: Drinking water during meals makes you gain weight.



FACT: Water has zero calories and cannot cause fat gain. It is a non-nutritive component.

2. The Building Blocks: Macro & Micro Nutrients

MACRONUTRIENTS: THE MAJOR PLAYERS



CARBOHYDRATES (Carbs)
Nutrients required in large amounts. They are the main part of your food.



PROTEIN
Building and repairing body tissues, muscles, nails, hair, cells.



FAT
Reserve energy, protects organs, regulates temp, helps absorb vitamins A, D, E, & K.



WATER
Transports nutrients, regulates temp, chemical reactions. Recommended 4-5 liters/day.

MICRONUTRIENTS: THE SUPPORTING CAST



VITAMINS

Nutrients required in small amounts, but vital for supporting the functions of macronutrients and overall health.



MINERALS

Inorganic elements, essential for body functions (e.g., Bone health: Calcium, Blood formation: Iron).

VITAMIN TYPES & DEFICIENCIES

Type	Description	Members	Mnemonic
FAT-SOLUBLE	Dissolve in fat, stored.	A, D, E, K	(Remaining)
WATER-SOLUBLE	Dissolve in water, not stored.	B-complex, C	"BC pani-pani ho gaya"

Vitamin	Scientific Name	Deficiency Disease
Vitamin A	Retinol	Night Blindness
Vitamin B1	Thiamine	Beriberi
Vitamin B3	Niacin	Pellagra
Vitamin B6	Pyridoxine	Anemia
Vitamin B7	Biotin	Dermatitis (Skin issues)
Vitamin C	Ascorbic Acid	Scurvy
Vitamin D	Calciferol	Rickets
Vitamin K	Phylloquinone	Excessive bleeding (no clotting)

4. Your Guide to a Healthy Weight

WHAT IS A HEALTHY WEIGHT?

Ideal weight for height; fit body, good posture, disease protection.

CALCULATE YOUR BMI

$$\text{BMI} = \frac{\text{WEIGHT (in kg)}}{[\text{HEIGHT (in m)}]^2}$$

UNDERSTANDING YOUR BMI SCORE



THE PITFALLS OF DIETING

Extreme calorie restriction is harmful. Leads to weakness, slower metabolism, nutrient deficiencies. Skipping meals is not the solution.

Made By Dr Pragya

6. Fueling for Victory: A Sports Diet Plan



2-3 hours before

PRE-COMPETITION DIET

Focus on high-carb diet for sustained energy. Consume easily digestible carbs, ensure good hydration.



Short breaks

DURING COMPETITION DIET

Replenish fluids, get instant energy. Bananas, glucose water, or carb-rich drinks.



1-2 hours after

POST-COMPETITION DIET

Focus on recovery. Consume protein-rich meal with carbs to repair muscle and restore energy.

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The Ultimate Guide to Fitness Tests: Class 12 Physical Education

FITNESS FUNDAMENTALS & SAI KHELO INDIA TEST (Ages 5-8)

What is Fitness?

The ability to perform any physical activity efficiently and effectively without feeling tired, while also maintaining overall health and well-being.

Test Group 1: Ages 5-8 (Classes 1-3)

These tests focus on Fundamental Movement Skills, which are the building blocks for physical activities like games, dance, and sports.

Test 1: Body Mass Index (BMI)



Purpose: To determine body composition (for one's muscle distribution).
Equipment: Weighing machine, measuring tape.
Formula: $\text{Weight (kg)} / (\text{Height (m)})^2$



Test 2: Plate Topping Test



Purpose: To measure limb coordination and speed.
Equipment: Table, two 20cm yellow disc, one 30x22cm rectangle, stopwatch.
Procedure: Tap the two discs 50 times (35 full cycles) as quickly as possible.

Test 3: Flamingo Balance Test



Purpose: To assess the ability to balance on a single leg.
Equipment: Non-slippery surface, stopwatch.
Procedure: Stand on one leg, holding the other foot, and maintain balance for 30 seconds.

SAI KHELO INDIA TEST (Ages 9-18)

Test Group 2: Ages 9-18 (Classes 4-12)

This battery of tests is designed to measure the overall physical fitness of older students.

Test 1: 50 Meter Speed Test



Purpose: To measure speed and acceleration.
Procedure: Sprint for 50 meters as fast as possible. The best of two trials is recorded.

Test 2: 600 Meter Run/Walk



Purpose: To measure cardiovascular fitness and endurance.
Procedure: Cover the 600m distance by running or fast-walking. Score is recorded in minutes and seconds.

Test 2: Sit & Reach Test



Purpose: To measure the flexibility of the lower back and hamstrings.
Equipment: Sit & Reach box.
Procedure: Sit with legs straight and reach forward as far as possible, holding for 5 seconds.

Test 4: Partial Curl-Up



Purpose: To measure abdominal strength and endurance.
Procedure: Perform as many curl-ups as possible until form breaks. Shoulders must raise at least 2 inches off the floor to count.

Test 3: Push-Ups (Boys) & Modified Push-Ups (Girls)



Purpose: To measure upper body strength and endurance.
Procedure (Boys): Standard push-up position.
Procedure (Girls): Modified position with knees on the ground.
Score: The total number of correct push-ups performed.

CARDIOVASCULAR FITNESS & METABOLISM

What is Cardiovascular Fitness?

How well your heart, lungs, and organs consume, transport, and utilize oxygen during long-duration activities.



The Harvard Step Test

Purpose: Measures cardiovascular fitness.
Procedure: Step up and down on a bench at a rate of 30 steps per minute for 5 minutes (or until exhaustion).

Harvard Step Test Equipment & Formula

Bench Height: 20 inches for males, 16 inches for females.
Fitness Index Score = $\frac{\text{Duration of exercise (sec)} \times 100}{\text{Pulse count (1-1.5 min post-exercise)}}$

What is Basal Metabolic Rate (BMR)?

The minimum number of calories your body needs to perform basic life-sustaining functions at rest.

BMR Multiplier Activity Levels

Sedentary	(Little to no exercise)	= BMR x 1.2
Light Activity	(Exercise 1-3 days/week)	= BMR x 1.275
Moderate Activity	(Exercise 3-4 days/week (30-60 mins))	= BMR x 1.55
Heavy Activity	(Exercise 5-7 days/week (1 hour))	= BMR x 1.725
Extreme Activity	(Body intense exercise or physical job)	= BMR x 1.9

RIKLI & JONES SENIOR CITIZEN FITNESS TEST

Designed for Seniors: These are modified, milder tests to assess functional fitness in older adults without causing strain.

Test 1: Chair Stand Test



Purpose: Lower body strength.
Procedure: Number of times a person can sit/stand up from a chair in 50 seconds with arms crossed.

Test 2: Arm Curl Test



Purpose: Upper body strength.
Procedure: Number of knee curls performed in 30 seconds.
Weight: 6 lbs for women, 8 lbs for men.

Test 3: Chair Sit & Reach Test



Purpose: Upper body flexibility.
Procedure: While sitting at the edge of a chair, reach forward the tip of one extended leg. Score is distance from toes (negative if short, positive if past).

Test 4: Back Scratch Test



Purpose: Upper body flexibility (shoulders).
Procedure: Reach one hand over the shoulder and the other up the middle of the back. Score is the distance between fingertips.

Test 5: Six-Minute Walk Test



Purpose: Aerobic endurance.
Procedure: The maximum distance a person can walk in 6 minutes around a marked area.

Test 6: 8-Foot Up-and-Go Test



Purpose: Speed, agility, and dynamic balance.
Procedure: Time taken to stand up from a chair, walk 8 feet, turn, and sit back down.

JOHNSON-METHANY TEST OF MOTOR EDUCABILITY

Purpose of the Test

To evaluate an athlete's general motor ability and neuro-muscular function through a series of four "motor stunts."

Test Area & Scoring

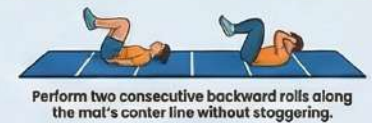
Performed on a 18ft x 2ft mat with 19 sections. Each successful stunt earns points (max 10 per stunt), with deductions for errors like staggering.

Stunt 1: Front Roll



Perform two consecutive forward rolls along the mat's center line without staggering.

Stunt 2: Back Roll



Perform two consecutive backward rolls along the mat's center line without staggering.

Stunt 3: Jumping Half Turn (180°)



Perform a series of four jumps, each with a half turn, skipping one mat line with each jump.

Stunt 4: Jumping Full Turn (360°)



Similar to the half turn, but executing a full 360° rotation with each jump while progressing down the mat.

Physiology & Injuries in Sports:

A Visual Guide for Class 12

THE PILLARS OF PHYSICAL FITNESS



What is Sports Physiology?

The study of how the human body functions and responds during exercise and sports activities to improve performance.

4 Key Physiological Factors Determining Fitness

Your performance in any sport is determined by these four core components.



Strength

The ability of muscles to exert force. Determined by muscle size, body weight, muscle composition (fast-twitch fibers), and the intensity of nerve impulses.



Speed

How quickly you can move. Depends on the mobility of the nervous system (reflexes), muscle composition (fast-twitch fibers), exclusive strength, and flexibility.



Endurance

The ability to sustain activity over time. Governed by aerobic capacity (oxygen use), lactate tolerance, movement economy, and muscle composition (slow-twitch fibers).



Flexibility

The range of motion in your joints. Influenced by muscle strength and stretchability, age, gender, and internal factors like body temperature (warming up is key).

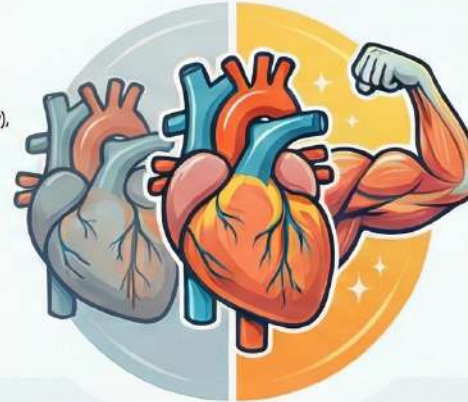
THE TRANSFORMATIVE EFFECTS OF EXERCISE

How Regular Exercise Changes Your Body

Consistent training leads to significant positive changes in the cardiorespiratory and muscular systems.

Cardiorespiratory System

- Stronger Heart**
Heart size increases (cardiac hypertrophy), improving its pumping capacity.
- Increased Efficiency**
Resting heart rate decreases, and stroke volume increases.
- Better Oxygen Transport**
Blood circulation and hemoglobin content improve.
- Enhanced Lung Capacity**
The ability to inhale and exhale air improves, along with gas exchange.



Muscular System

- Muscle Growth**
Muscle size and strength increase (muscle hypertrophy).
- Delayed Fatigue**
Endurance increases due to less lactic acid accumulation.
- Improved Posture**
Muscles become more toned and provide better support.
- Reduced Injury Risk**
Flexibility and elasticity of muscles increase.



UNDERSTANDING SPORTS INJURIES

Common Injuries in Sports

Injuries are broadly classified into three types: soft tissue, bone, and joint injuries.

Soft Tissue Injuries (Muscles & Ligaments)



- Strain**
A pulling or tearing of a muscle or tendon.
- Sprain**
An overstretching or tearing of a ligament.
- Contusion & Abrasion**
Contusion is a bruise from a direct impact, while abrasion is a scrape caused by friction.

First Aid for Strains & Sprains



RICE Method (Traditional)

Rest, Ice, Compression, Elevation. Focuses on stopping activity immediately.



MICE Method (Modern)

Movement (gentle), Ice, Compression, Elevation. Aims to prevent stiffness.

Bone Injuries (Fractures)



Greenstick
An incomplete fracture where the bone bends and cracks, common in children.



Comminuted
The bone shatters into three or more pieces.



Transverse
A straight break across the bone at a 90° angle.



Oblique & Impacted
Oblique is a diagonal break. Impacted is when one broken end is driven into the other.

Joint Injuries (Dislocation)



Dislocation
Occurs when a bone is forced out of its normal position in a joint (e.g., shoulder, hip, knee).

PHYSIOLOGICAL CHANGES DUE TO AGING



- Muscular System Weakens**
Muscle strength, size, and flexibility naturally decrease over time.
- Skeletal System Becomes Brittle**
Bone density decreases, leading to conditions like osteoporosis and joint stiffness.
- Cardiorespiratory Efficiency Declines**
The functional capacity of the heart and lungs is reduced.
- Nervous System Slows**
Reflexes become slower, coordination may be affected, and memory can decline.

THE SCIENCE OF SPORT: YOUR GUIDE TO BIOMECHANICS



WHAT IS BIOMECHANICS?

Bio (Life) + Mechanics (Movement).
Biomechanics is the science that studies the movement of a living body, including how muscles, bones, and ligaments work together.



Analyzing Sports Movements:
Studying how a bowler's shoulder rotates or how a long jumper generates maximum force from their legs using gravity and friction.



THE 3 CLASSES OF LEVERS IN THE HUMAN BODY

How Levers Work: A lever is a rigid structure (bone) that moves around a fixed point called a fulcrum (joint), with force applied by muscles to move a load.

CLASS 1 LEVER: FULCRUM IN THE MIDDLE



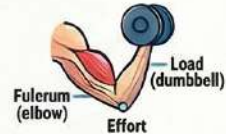
The fulcrum is positioned between the effort (force) and the load. As a classic example is a seesaw.

CLASS 2 LEVER: LOAD IN THE MIDDLE



The load is positioned between the fulcrum and the effort. (e.g., Performing a calf raise by standing on your toes).

CLASS 3 LEVER: EFFORT IN THE MIDDLE



The effort is applied between the fulcrum and the load. (e.g., A bicep curl, where the elbow is the fulcrum and the bicep muscle provides the effort).

FRICTION AND ITS ROLE IN SPORTS

What is Friction? Friction is a force that opposes motion when one surface moves over another. It's why a rolling ball eventually stops.



Static Friction
The friction on an object that prevents it from moving when a force is applied.



Sliding & Rolling Friction
Sliding friction occurs when objects slide over each other. Rolling friction (e.g., pushing a very heavy, immovable object).



Fluid Friction (Air/Water Resistance)
The resistance experienced when moving through a fluid. (e.g., Air resistance pushing against a runner, or water resistance against a swimmer).



PROJECTILE MOTION

The Path of a Thrown Object:
When an object is thrown into the air, it follows a curved, parabolic path due to the combined effects of horizontal motion, gravity, and air resistance.

45°

The Optimal Angle for Distance:
To achieve the maximum horizontal distance, an object should ideally be projected at a 45° angle.

Key Factors Affecting a Projectile:
Speed of release, height of release, air resistance, and gravity all influence the object's path.

Example:



Projectiles in Every Sport:
Examples include a javelin throw, a shot put, a kicked football, a basketball shot, and even the human body during a long jump.

NEWTON'S LAWS OF MOTION IN SPORTS

1ST LAW: LAW OF INERTIA

An object will stay at rest or continue moving in a straight line unless an external force acts on it. (e.g., A football remains still until kicked).



Inertia in Action: This law is applied in sports like the hammer throw, lifting an opponent in wrestling, and a sprinter's start from the blocks.

2ND LAW: LAW OF ACCELERATION ($F = ma$)

The acceleration of an object is directly proportional to the force applied and inversely proportional to its mass.

More Force = More Speed:
To throw a cricket ball fast or a shot out for, you must apply a large force to generate high acceleration.

3RD LAW: LAW OF REACTION

For every action, there is an equal and opposite reaction.



Action & Reaction in Sports:
In swimming, you push water backward (action) and the water pushes you forward (reaction). In long jump, an athlete pushes down on the ground to be propelled upward.

BALANCE & STABILITY

Equilibrium: The State of Balance:
Equilibrium is a condition where opposing forces are balanced, resulting in a stable state.

Static vs. Dynamic Equilibrium



Static is balance while stationary (standing, sitting).



Dynamic is balance while in motion (running, gymnastics).



Center of Gravity (COG):
An imaginary point where the entire weight of the body is considered to be concentrated and balanced.

COG Affects Stability:
The COG's position changes with movement. A lower center of gravity generally leads to greater stability, which is crucial for performing sports stunts correctly.

Psychology & Sports: A Class 12 Quick Guide

WHAT IS PERSONALITY?

No Single, Specific Definition

Personality is best understood as the characteristic integration of an individual's inner and outer qualities, including their physical, mental, and emotional traits.

It's How We Think, Behave & Interact

It is the combination of a person's patterns of thinking, their behavior, and their ability to handle emotions and interact with others.

CLASSIFYING PERSONALITY TYPES

Jung's Classification



Introvert
This theory classifies personality into three types based on mental and social characteristics.

Introverts are reserved and prefer solitude. Extroverts are outgoing and social. Ambiverts are a balanced blend of both, often considered the ideal type.

The Big Five Theory (OCEAN)



A-Agreeableness Cooperative, kind, and trusting; can be easily influenced. Example: Masao
C-Conscientiousness Example: Dekisugi
E-Extraversion Example: Gian
N-Neuroticism Example: Nobita
O-Openness Example: Shizuka

This model identifies five core personality traits that exist on a spectrum. An easy way to remember them is with the acronym OCEAN.

MOTIVATION: THE DRIVE TO ACT

What is Motivation?

The internal energy or reason that stimulates an individual to take action to achieve a desired goal.

Intrinsic Motivation

Intrinsic motivation comes from within (self-satisfaction).

Intrinsic vs. Extrinsic



Intrinsic Motivation is More Sustainable:
Self-motivation is crucial for long-term success, as extrinsic motivation can fade over time.

Extrinsic Motivation

Extrinsic motivation comes from external rewards (prizes, praise).



Ways to Motivate
• Techniques include setting clear goals, offering each prizes or trophies, verbal encouragement, motivational music, and positive self-talk.

EXERCISE ADHERENCE

Sticking to the Plan

Exercise adherence is the ability to consistently follow an exercise program over a long period to achieve your fitness goals.



Reasons & Benefits

People exercise to improve their looks, boost self-confidence, reduce stress, and improve sleep. This leads to better mental health, stronger bones, and a positive body image.



Strategies for Success
To improve adherence, have a clear reason for your workout, maintain a positive self-perception, and seek positive feedback and motivation from a trainer.

KEY PSYCHOLOGICAL ATTRIBUTES FOR ATHLETES

Self-Esteem

An individual's overall sense of self-worth and confidence in their own abilities, which leads to better decision-making.



Self-Talk

The internal conversation you have with yourself. Positive self-talk can replace negative thoughts and improve focus.

Mental Imagery

The skill of using your imagination to visualize yourself performing successfully, helping with strategy and confidence.

Goal Setting

The process of identifying what you want to achieve and creating a plan to get there, which provides motivation and direction.

THE 4 DIMENSIONS OF PERSONALITY

1. Physical Dimension



1. Physical
Based on physical appearance ('First impression is the last impression'), but doesn't define it entirely.

2. Mental
Considered the 'root' phenomenon of personality. It's the foundation of personality.

3. Social Dimension
Reflects how a person interacts with others. Key qualities include kindness, helpfulness, leadership, and good manners.

4. Emotional



4. Emotional
The emotion of the person and the emotion under store to get doesn't define it entirely.

AGGRESSION IN SPORTS

Behavior Intended to Harm

Aggression in sports is any behavior aimed at harming an opponent either physically or psychologically.



1. Hostile Aggression

A direct act of violence or harm towards another person, driven by anger (e.g., physically fighting an opponent).

2. Instrumental Aggression

An aggressive act that is a means to achieve a specific goal, not just to cause harm (e.g., being aggressive within the rules to prove you are the best).

3. Assertive Behavior

Using legitimate force without intent to harm. In sports, this can also refer to provoking an opponent (e.g., through sledging) to make a mistake.

THE ULTIMATE GUIDE TO TRAINING IN SPORTS

COMPONENT 1: STRENGTH

TWO MAIN TYPES OF STRENGTH



Static Strength (Isometric)
The ability to exert force without any visible movement of the body part, such as holding a heavy weight in a fixed position.



Dynamic Strength (Isotonic)
The ability to exert force with repeated movements.
Builds up:
• Muscular Strength
• Explosive Strength
• Strength Endurance

WHAT IS STRENGTH?
The ability of a muscle or group of muscles to exert force against resistance in a single maximal effort.

HOW TO IMPROVE STRENGTH

Isometric Exercises
(e.g., Planks)

Isotonic Exercises
(e.g., Push ups)

Isokinetic Exercises
(e.g., Cycling machine)

COMPONENT 3: SPEED

TYPES OF SPEED ABILITIES



Reaction Time



Acceleration Ability



Speed of Movement



Locomotor Ability

WHAT IS SPEED?
The ability to achieve a maximum of cover a distance in the shortest possible time.

HOW TO IMPROVE SPEED

Acceleration Runs
Sprinting at maximum effort over a short distance (20-30 meters) for 5-10 repetitions with sufficient rest in between.

Pace Runs
Running a longer distance (e.g., 800 meters) at a normal pace and then finishing the last 20% of the distance at maximum speed.

THE FOUNDATIONS OF TRAINING



WHAT IS SPORTS TRAINING?
A special process of preparation for an athletes that uses scientific principles to improve and maintain performance for sporting activities.

KEY FACTORS IN TALENT IDENTIFICATION



Physical Attributes



Physiological Traits



Technical Skills



Psychological Strength



Social Skills



WHAT IS TALENT IDENTIFICATION?
A process of recognizing promising athletes at regional or social levels and providing them with support to become national or international champions.



TALENT DEVELOPMENT IS CRUCIAL
After identifying talent, it is to develop and refine them to the long term to ensure they full potential.



Cognitive Skills

PERIODIZATION: STRUCTURING YOUR TRAINING

WHAT IS PERIODIZATION?
An organized approach that divides the overall training into smaller, strategic periods or cycles to achieve specific goals.



1. MACRO CYCLE
The longest training period, typically lasting an entire season. It encompasses all phases of training, preparation, competition, and maximum.

2. MESO CYCLE
A training block within the Macro Cycle that lasts for several weeks to a few months, focusing on specific goals like improving speed or strength.

3. MICRO CYCLE
The shortest training period, lasting from a few days to a week. It consists of specific daily workouts and training sessions.



COMPONENT 2: ENDURANCE

WHAT IS ENDURANCE?
The ability to sustain an activity for a long period without getting tired or bloated. Synonymous with stamina.

TYPES OF ENDURANCE



Speed Endurance
Performing an activity at a high-speed intensity for an extended time.



Strength Endurance
The ability to overcome endurance for a long duration.



Long-Term Endurance
Performing an activity for a very long duration at a normal intensity.

HOW TO IMPROVE ENDURANCE

Continuous Training
Running without breaks lasting without breaks.

Continuous Training
Performing an exercise like running at a slow pace for a long duration (1-2 hours) without any breaks. Can be slow, fast, or variable pace.



Fartlek Training
A varied method meaning "speed play". It involves continuous training over various terrains, using various exercises.



COMPONENT 4: FLEXIBILITY

WHAT IS FLEXIBILITY?
The range of movement possible at a specific joint. Greater flexibility helps prevent injuries and improves posture.

TYPES OF FLEXIBILITY



Active Flexibility
Achieving a stretched position without any external help.



Passive Flexibility
Achieving a stretched position with the help of an external partner or equipment.

HOW TO IMPROVE FLEXIBILITY

Ballistic Method
Using swinging or bouncing movements to stretch muscles.

Static Stretching
Slowly stretching to a final position and holding it for a period of time.

PNF Technique
An advanced technique, often done with a partner, to achieve a greater range of motion.



COMPONENT 5: COORDINATIVE ABILITIES & CIRCUIT TRAINING

WHAT ARE COORDINATIVE ABILITIES?
The ability to perform movements with perfection and efficiency. Includes: Balance Ability, Distance Ability, Reaction Ability, and Coupling Ability.

WHAT IS CIRCUIT TRAINING?

A training method where you perform a series of different exercises one after another with minimal rest, targeting different body parts in a short time.



BENEFITS OF CIRCUIT TRAINING
Improves physical endurance and strength, time efficient, activates multiple muscle groups, and enhances cardiovascular fitness.

DEVELOPED BY ADAM RUIZ AND MORRAN:
The popular training method was first introduced in 1967.